AI HACKATHON BY JASECI · TEAM GENIE

lumira

From floor plan to power plan.

AI-assisted electrical design for Sri Lankan homes



THE PROBLEM

Most homes are still wired from a sketch.

Hand-drawn layouts

Homeowners rarely see where lights and sockets go until the walls are open.

Rule-of-thumb counts

Light and socket numbers come from habit, not a lux target or the room's size.

No load check

Few people add up the whole house against a 30 A single-phase main switch.

WHO IT’S FOR

Everyone who plans a home.

Homeowners

See the layout and the load before work starts.

Architects

Lay the electrical points on the same plan.

Electricians

Start from a clear, labelled first draft.

Engineers

Review it: every value names its rule and source.

Upload a plan. Get a rule-based first draft of the wiring.

01

Import any plan

02

Suggest wiring

03

Check the load

AI reads the rooms, and asks when unsure.

Any plan in

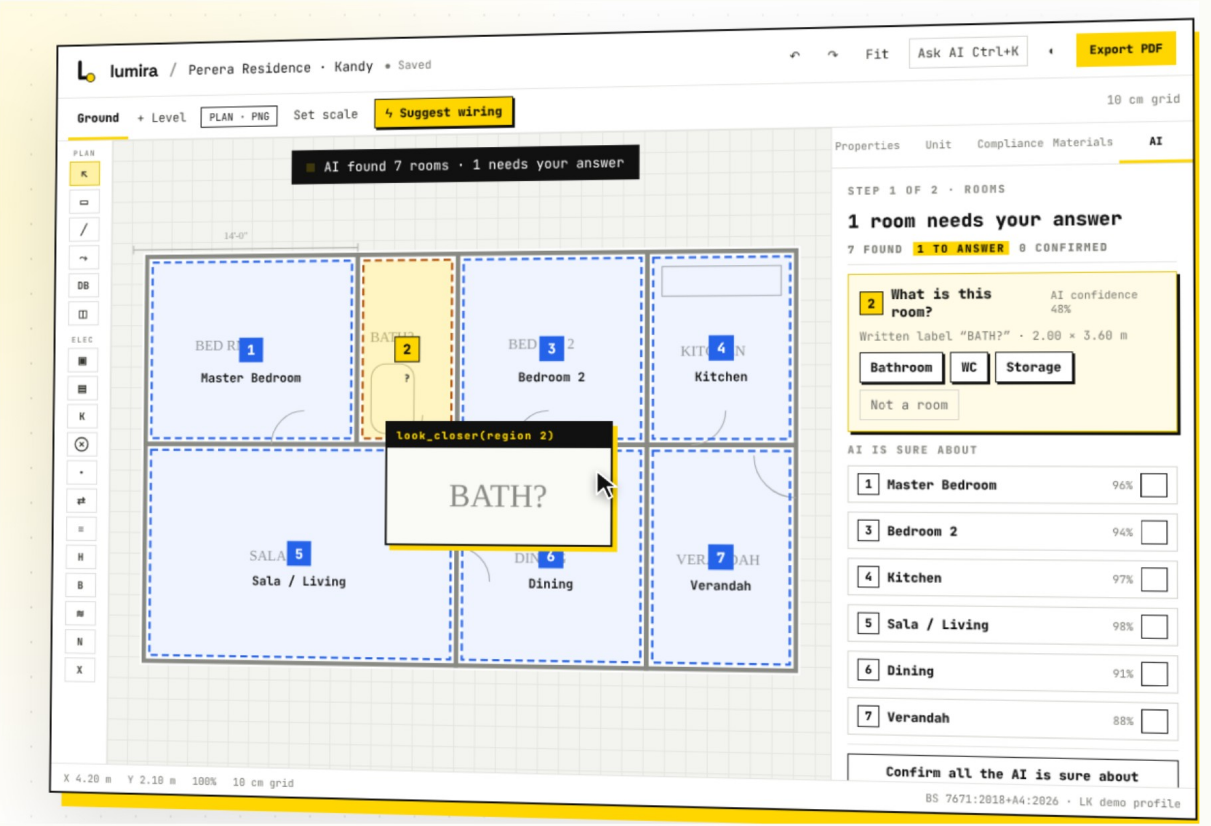
PNG, JPG, PDF, SVG or DXF. Two clicks set the scale.

OpenCV finds rooms

Thick strokes are walls; text and door swings never split a room.

An AI agent names them

It zooms in on small labels and asks you about low-confidence rooms.



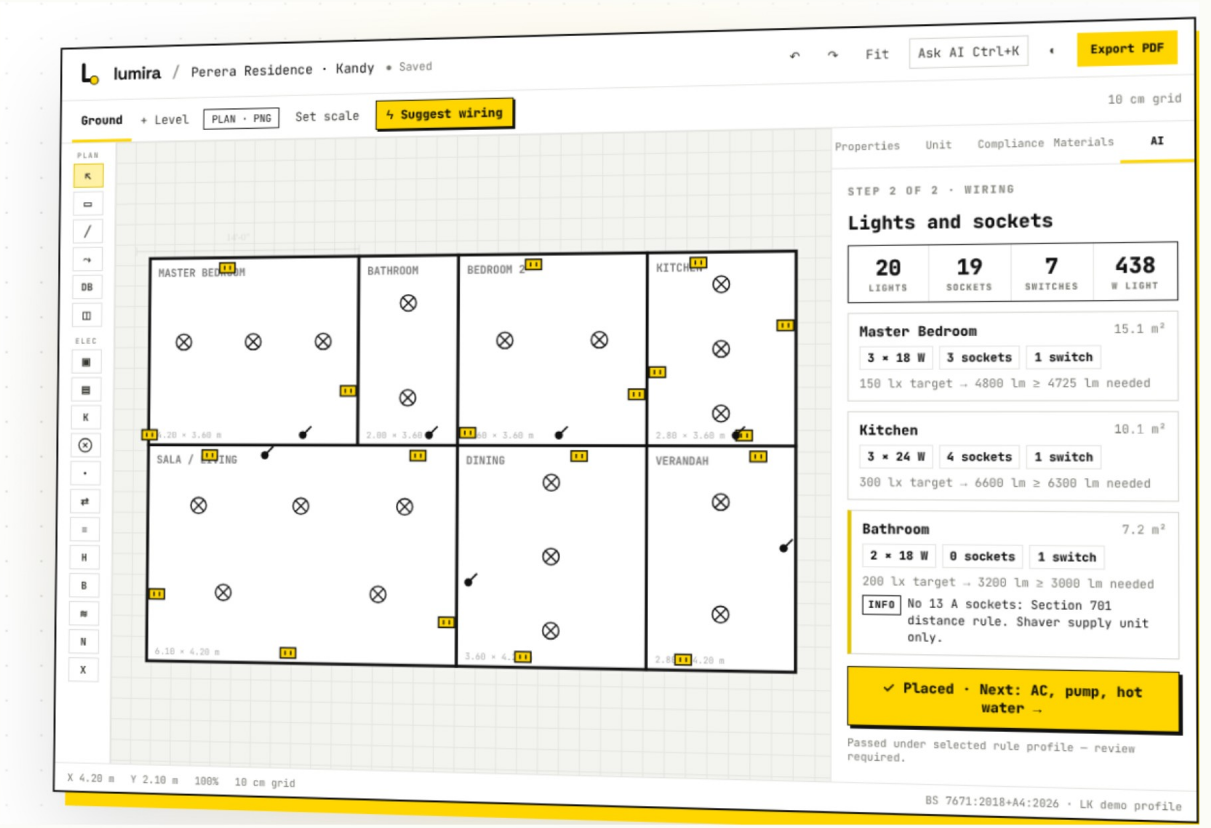
Lights and sockets, calculated.

MASTER BEDROOM · 15.1 m²

150 lx × 15.1 m² ÷ (0.6 × 0.8)
= 4,725 lm needed

3 × 18 W LED

A switch beside every door. Sockets scale with room size, and none in bathrooms (Section 701). Preview, then place.



Catches an overloaded house early.

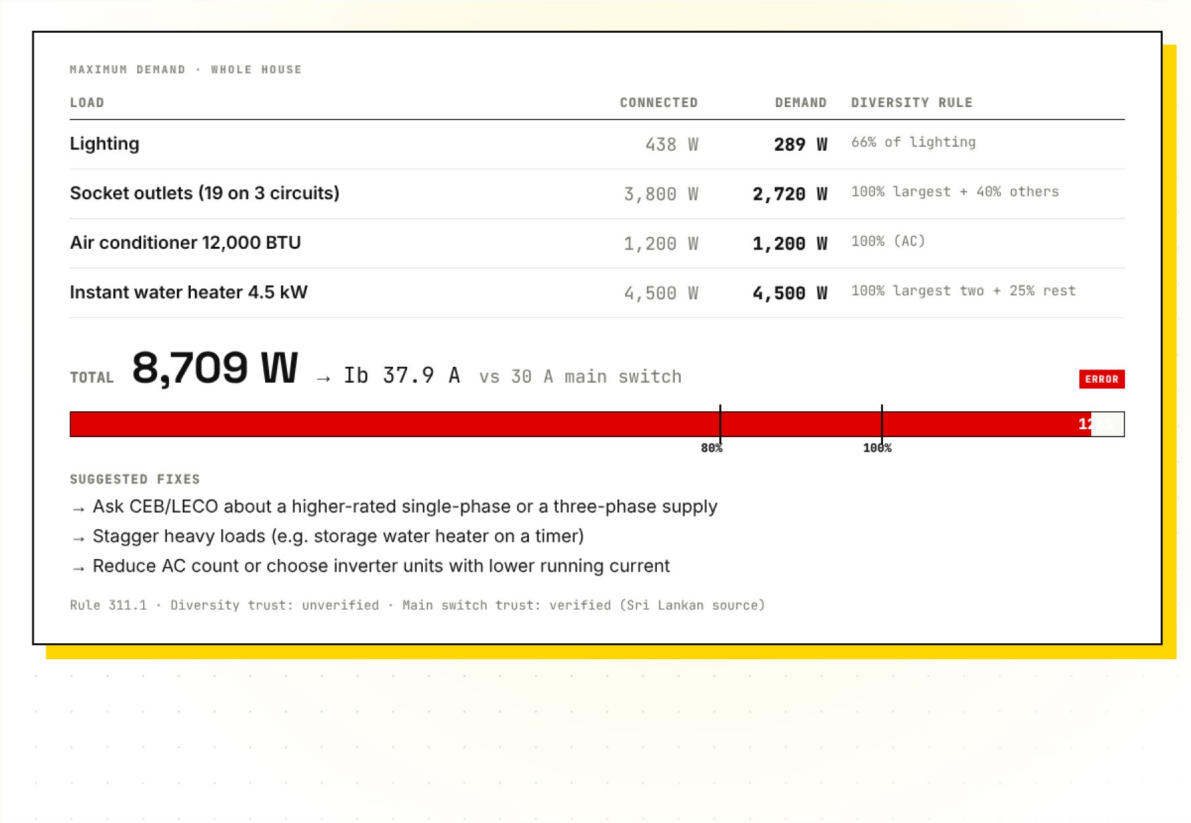
AC sized from the room

15.1 m² × 700 BTU/m² → a 12,000 BTU unit on B20, 2.5 mm².

Whole-house demand

8,709 W with diversity = 37.9 A on a 30 A main switch.

126% · ERROR



AI names. Code calculates.

CODE	CODE	AI	YOU	CODE	CODE
Import	**Find rooms**	**Name rooms**	**Confirm**	**Layout**	**Loads**
PyMuPDF · ezdxf	OpenCV	by llm()	human check	lumen method	rule profile

Jac graph: **Project** → **Level** → **Room** → **Device**

Every number names its rule and its source.

BS 7671:2018+A4:2026

Adapted for Sri Lanka: 230 V, 50 Hz, TT earthing, a 30 A single-phase main switch.

Never says “safe” or “compliant”: **“Passed under selected rule profile — review required.”**

Trust tag	Means
verified_current	Official, current source
verified_2008_guide	Concept OK, value may have moved
secondary	Quotes BS 7671, not official
unverified	Model knowledge: check it

Three things we had to get right.

Messy plans

Door gaps merged rooms and dimension lines split them, until we separated thick walls from thin lines.

An honest AI

Handwritten labels fool it, so it reports confidence and asks instead of guessing.

Real-world diversity

Rating-based socket diversity failed every 30 A house; we use estimated load per outlet.

WHAT'S NEXT

From first draft to full design pack.

Circuits & DB schedule

Group points into circuits with a consumer unit layout.

Voltage-drop checks

Per-circuit checks using real cable runs.

Materials in LKR

A bill of materials with local prices.

Natural-language edits

"Add an AC to every bedroom" in the Ask AI box.

PDF export

Drawings ready for the electrician.

Verified rules

Rule values checked with chartered engineers.



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`github.com/RanaweeraYHGR/wirewise-ai-editor`

Thank you · Questions? · Preliminary designs, always reviewed by a qualified electrical professional.